Statistical and Machine Learning Models

STAT 525 - Topics in Computational Statistics

Shulei Wang

University of Illinois at Urbana-Champaign

shuleiw@illinois.edu

https://sites.google.com/view/shulei-wang/



Overview

Introduction

Statistical and Machine Learning Models

What is this course about?

- Many statistical and machine learning methods can be formulated as an optimization problem (minimize the loss function)

$$\min f(x) \qquad \text{s.t.} \quad x \in \Omega.$$

  - (*algorithm*) How can we find the minimum efficiently and robustly?
  - (*computation complexity*) How many iterations/query points are sufficient to find the minimum?
- This is a relatively theoretical course and it needs some mathmatical backgrounds.

How we learn this course

- Instructor: Shulei Wang
  - Assistant Professor of Statistics, University of Illinois at Urbana-Champaign
  - Office hour: Tuesday 9:00 am - 10:00 am (CST)
  - Email: shuleiw at illinois dot edu
- TA: Zhiyu Wang
  - Office hour: Tuesday and Thursday 2:00 pm - 3:00 pm (CST)
  - Email: zhiyuw6 at illinois dot edu

How we learn this course

- What you need to do every week
  - Check our course website on Canvas
  - Come to lectures (2-3 hours)
  - Read the materials (3-5 hours)
  - Ask questions during office hour
  - Finish bi-weekly homework (1-2 hours)

How we learn this course

- Prerequisite:
  - Linear algebra (e.g. Math 415)
  - Some analysis (e.g. Math 444)
  - Some statistics and machine learning background (e.g. Stat 527)
- Grades:
  - Bi-weekly homework assignments (60%): The lowest score will be dropped.
  - Final project (40%): The students need to give a presentation in the last two weeks of the semester.

Books

- Optimization for Data Analysis, by *Wright & Recht* (Recommended)
- First-order and Stochastic Optimization Methods for Machine Learning, by *Lan* (Optional)
- Convex Optimization, by *Boyd & Vandenberghe* (Optional)
- Numerical Optimization, by *Nocedal & Wright* (Optional)
- First-Order Methods in Optimization, by *Beck* (Optional)
- Lectures on Convex Optimization, by *Nesterov* (Optional)
- High-Dimensional Data Analysis with Low-Dimensional Models: Principles, Computation, and Applications, by *Wright & Ma* (Optional)

Topics

1. Statistical and Machine Learning Models
2. Convex Optimization Theory
3. Deterministic Smooth Convex Optimization
4. Deterministic Non-smooth Convex Optimization
5. Stochastic Optimization

Questions

Any questions?

Introduction

Statistical and Machine Learning Models

Statistical and Machine Learning Models

- We will review several popular statistical and machine learning models and show why we are interested in solving optimization problems.
- We won't dig into details of these models. If you are interested in such models, read the book
  - "Elements of Statistical Learning" by *Hastie, Tibshirani & Friedman*
  - "High-Dimensional Data Analysis with Low-Dimensional Models: Principles, Computation, and Applications" by *Wright & Ma*
  - "Optimization for Data Analysis" by *Wright & Recht*

Statistical and Machine Learning Models

- In most supervised learning problem, we collect data

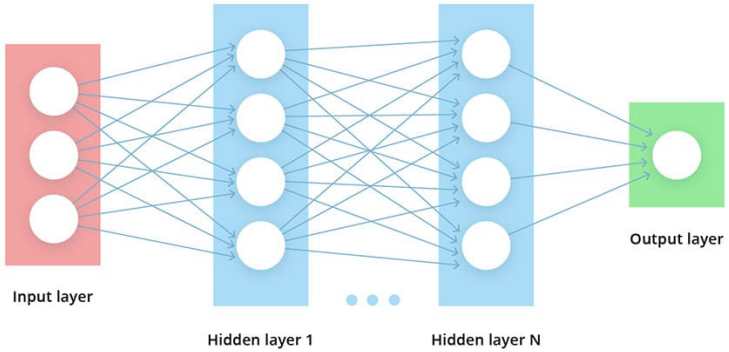
$$\mathcal{D} = \{(X_i, Y_i), 1 \leq i \leq n\},$$

  and are interested in learning a function ϕ_β such that $\phi_\beta(X_i) \approx Y_i$.
- In these problems, we can find the best ϕ_β by considering the following optimization problem

$$\min_{\beta \in \mathcal{B}} \underbrace{\frac{1}{n} \sum_{i=1}^{n} \mathcal{L}(\phi_\beta(X_i), Y_i)}_{\textit{Loss}} + \underbrace{\lambda \mathcal{R}(\beta)}_{\textit{Regularizer}}$$

- We consider several examples.

Deep Learning



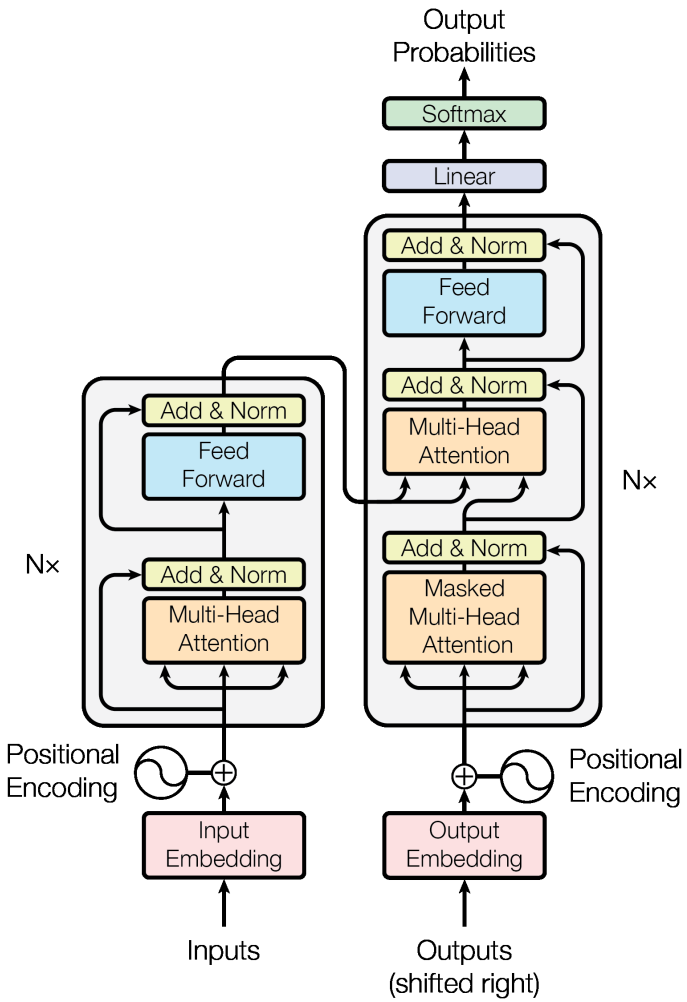
- In recent years, deep neural networks emerges as a powerful nonlinear classifier. The transformation at layer l

$$X_i^l = \sigma\left(W^l X_i^{l-1} + g^l\right),$$

where σ is called an activation function, including

  - Sigmoid: $\sigma(t) = 1/(1+e^{-t})$
  - Rectified Linear Unit: $\sigma(t) = \max(t, 0)$

Deep Learning



- Sometimes, the architecture of deep neural networks can be complicated.

Deep Learning

- In classification, softmax is applied to the last layer of deep neural networks so we can have "probability" as output.
- Cross entropy loss is usually used

$$\min_{\beta} \frac{1}{n} \sum_{i=1}^{n} \left[\sum_{k=1}^{K} Y_{i,k} \psi_\beta(X_i)_k - \log \left(\sum_{k=1}^{K} \exp(\psi_\beta(X_i)_k) \right) \right].$$

  Here, ψ_β is deep neural networks with parameter β.
- The loss function is usually nonconvex in β.

Optimization

- In all statistical and machine learning models, we can formulate them as optimization problems and also need to know how to solve them.
- In modern statistics and machine learning, optimization plays a very important role.
- After this class, we won't talk about these specific models and will focus on optimization problem alone.
- Hopefully, you will learn how to efficiently solve your own optimization problem in your statistical model after this course.

First Order Optimization Methods

- The optimization problem can always be written as

$$\min_{x \in \Omega} f(x).$$

- In this course, we will focus on the first order methods, i.e. only need the derivative information of $f(x)$, e.g.

$$x_{t+1} = x_t - \alpha_t \nabla f(x_t).$$

- We won't talk about the second order methods, e.g. Newton method and Quasi-Newton method.